CHAPTER 6

CLIENT SIDE SCRIPTING USING JAVASCRIPT

CHAPTER ESSENCE AT A GLANCE:
Implement client-side interactive logic, event handling, control structures, and DOM form validation using modern JavaScript functions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Dynamic Scripting in the Browser:** JavaScript provides a lightweight, interpreted, object-based scripting language executed directly within web browsers to transform static markup documents into responsive, interactive user experiences.

• **Variable Typing and Equality Semantics:** Dynamic variable typing adapts automatically to stored values; standard equality performs automatic type coercion, whereas strict identity requires both identical data values and identical underlying data types.

• **Document Object Model Manipulation:** The browser constructs an in-memory document tree where each element is an accessible object node, enabling scripts to read, modify, style, or generate content dynamically.

• **Interactive Dialog and Feedback:** Built-in browser dialog interfaces provide synchronous interaction by presenting alert notices, requiring confirmation booleans, or collecting direct textual user inputs.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$(x == y) \rightarrow \text{Value Equality} \quad | \quad (x === y) \rightarrow \text{Strict Value + Type}$

JavaScript standard versus strict equality comparison

$$\text{element} = \text{document.getElementById}(\text{"inputID"})$$

Document Object Model unique element node selection

$$\text{alert}() \rightarrow \text{void} \quad | \quad \text{confirm}() \rightarrow \text{boolean} \quad | \quad \text{prompt}() \rightarrow \text{string}$$

Built-in browser dialog function return signatures

$$\text{Average Value} = \frac{\sum_{i=1}^{n} \text{Score}_i}{N}$$

Array aggregation and statistical evaluation formula

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** JavaScript is strictly case-sensitive; variable names like totalScore and TotalScore represent distinct identifiers.

• **Alert:** Use isNaN() to validate numeric input values before performing arithmetic calculations on text field values.

• **Alert:** Return false in onsubmit handler functions to prevent form submission if validation errors are detected.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is JavaScript? How is JavaScript code embedded into an HTML document using <script> tags? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Definition):** Lightweight, interpreted, dynamic, object-based client-side scripting language used to create interactive web pages. **[1 Mark]**

• **Step 2 (Embedding in HTML):** <script type='text/javascript'>n alert('Hello Kerala HSE!');n</script>nOr via external file: <script src='script.js'></script>. Placed in <head> or <body>. **[1 Mark]**

• **Key Exam Result:** Embedded using <script> tag in <head> or <body>

$$(x == y) \rightarrow \text{Value Equality} \quad | \quad (x === y) \rightarrow \text{Strict Value + Type}$$

JavaScript standard versus strict equality comparison

2. Explain primitive and non-primitive data types in JavaScript. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Data Types):** 1. Number: Integers and floats (e.g. 25, 98.6). 2. String: Text in quotes (e.g. 'Kerala', '123'). 3. Boolean: true or false. 4. Null: Intentional absence of value. 5. Undefined: Variable declared but not assigned a value. **[2 Marks]**

• **Key Exam Result:** Primitive: Number, String, Boolean, Null, Undefined

$$(x == y) \rightarrow \text{Value Equality} \quad | \quad (x === y) \rightarrow \text{Strict Value + Type}$$

JavaScript standard versus strict equality comparison

3. Explain operators in JavaScript: Arithmetic, Relational, Logical, and Assignment operators. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Operators):** Arithmetic: +, -, *, divided by , % (modulus), ++, --. Relational: ==, != , === (strict equal with type), >, <, >= , <= . Logical: && (AND), || (OR), ! (NOT). Assignment: =, +=, -=, *=, divided by =. **[2 Marks]**

• **Key Exam Result:** Arithmetic, Relational (===), Logical, Assignment

$$(x == y) \rightarrow \text{Value Equality} \quad | \quad (x === y) \rightarrow \text{Strict Value + Type}$$

JavaScript standard versus strict equality comparison

4. Explain built-in dialog boxes in JavaScript: alert(), confirm(), and prompt(). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Dialog Boxes):** alert('Message'): Displays alert with OK button; returns undefined. confirm('Are you sure?'): Displays message with OK/Cancel; returns true (OK) or false (Cancel). prompt('Enter name:', 'Default'): Input dialog box; returns entered string or null. **[3 Marks]**

• **Key Exam Result:** alert() (notification), confirm() (yes/no), prompt() (input)

$$\text{alert}() \rightarrow \text{void} \quad | \quad \text{confirm}() \rightarrow \text{boolean} \quad | \quad \text{prompt}() \rightarrow \text{string}$$

Built-in browser dialog function return signatures

5. Explain control structures in JavaScript: if-else, switch, while, do-while, and for loops. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Control Structures):** Branching: if, if-else, switch-case. Looping: while (entry-controlled), do-while (exit-controlled), for (counter-controlled). break terminates loop; continue skips current iteration. **[3 Marks]**

• **Key Exam Result:** if-else, switch, while, do-while, for loops

$$\text{Average Value} = \frac{\sum_{i=1}^{n} \text{Score}_i}{N}$$

Array aggregation and statistical evaluation formula

6. Explain built-in functions in JavaScript: isNaN(), parseInt(), and parseFloat(). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Built-in Functions):** 1. isNaN(val): Returns true if value is Not-a-Number (e.g. isNaN('abc') is true, isNaN(123) is false). 2. parseInt(str): Parses string and returns an integer (parseInt('45.8') returns 45). 3. parseFloat(str): Parses string and returns floating point number (parseFloat('45.8') returns 45.8). **[3 Marks]**

• **Key Exam Result:** isNaN(), parseInt(), parseFloat()

$$\text{Average Value} = \frac{\sum_{i=1}^{n} \text{Score}_i}{N}$$

Array aggregation and statistical evaluation formula

7. Explain the Document Object Model (DOM) and how elements are accessed using document.getElementById(). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (DOM Concept):** Document Object Model represents HTML document as a hierarchical tree of objects accessible by scripts. **[1 Mark]**

• **Step 2 (Methods):** document.getElementById('uid'): Selects element by ID. document.getElementsByTagName('p'): Selects elements by tag. Element properties: innerHTML, value, style.color. **[2 Marks]**

• **Key Exam Result:** DOM: tree of HTML nodes; document.getElementById()

$$\text{element} = \text{document. getElementById}(\text{"inputID"})$$

Document Object Model unique element node selection